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Assignment 2

Optical Coherence Tomography Imaging System

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1 Introduction

Optical Coherence Tomography (OCT) is a non-invasive medical imaging method that uses near-infrared light to produce cross-sectional images of biological tissue. It is often described as an optical version of ultrasound because both methods form images from waves reflected or backscattered by tissue. The difference is that OCT uses light instead of sound, so it can show much finer details at the micrometre scale [1, 2].

OCT is most widely used in ophthalmology because the eye allows light to pass through and return useful structural information. It can show the retina, macula, optic nerve head, cornea, and anterior segment in detail. This makes it useful for detecting and monitoring conditions such as glaucoma, diabetic macular oedema, and age-related macular degeneration [3].

The main advantage of OCT is that it gives depth-resolved tissue information without cutting or removing tissue. It can also be repeated without ionising radiation exposure, which makes it useful for follow-up imaging. However, OCT has limited penetration depth because light is scattered and absorbed in tissue. Therefore, it is mainly useful for transparent or superficial structures, and the image quality depends strongly on good acquisition technique.

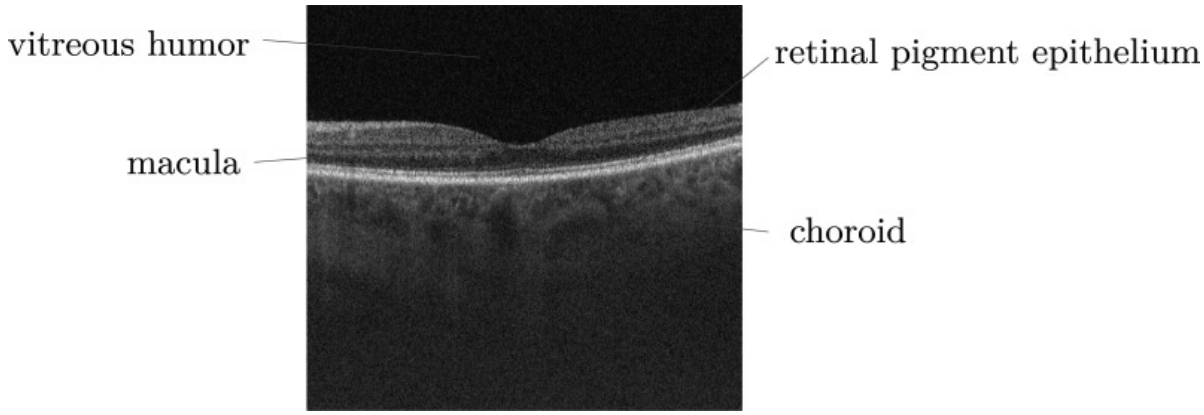


Figure 1: Example of a retinal OCT B-scan showing depth-resolved retinal structure. Brighter regions represent stronger optical backscattering from tissue layers. Adapted from [2].

2 Principles of Operation

2.1 Low-Coherence Interferometry

The basic principle of OCT is low-coherence interferometry. Light from a broadband source is split into two paths: a sample arm and a reference arm. The sample beam enters the tissue and is partly backscattered from different depths, while the reference beam travels to a mirror with a known path length.

When the returning sample light and reference light are recombined, interference occurs only when their optical path lengths are nearly matched. This allows the system to identify the depth from which the reflected light came. In simple terms, OCT measures the echo delay of light, but since light travels too fast to measure the delay directly, interference with a reference beam is used instead [2].

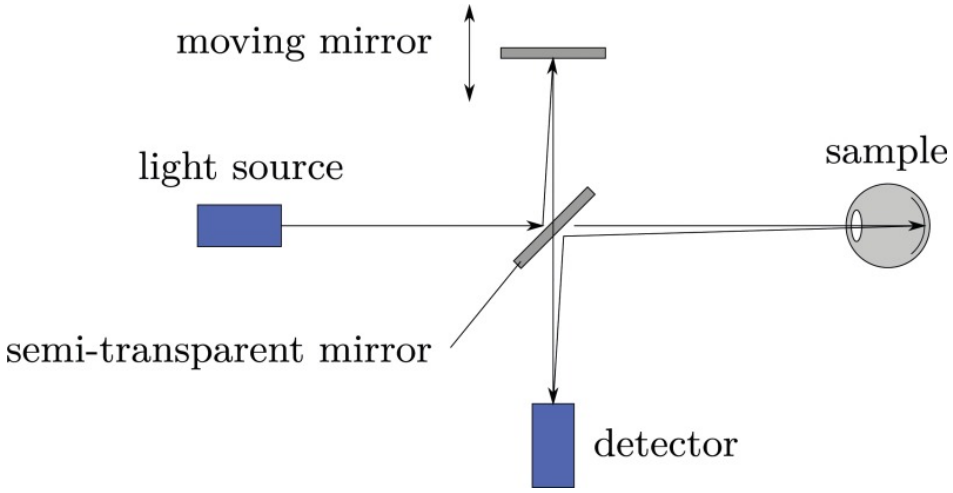


Figure 2: Basic OCT system based on low-coherence interferometry. Light is divided into a sample arm and a reference arm, and the returning beams are recombined to obtain depth information. Adapted from [2].

2.2 A-scans, B-scans, and Volume Imaging

An A-scan is the simplest OCT signal. It is a one-dimensional depth profile that shows how strongly light is reflected from different depths at one point on the tissue. When the OCT beam is moved sideways and many A-scans are collected, they are placed side by side to form a two-dimensional cross-sectional image called a B-scan.

For 3D imaging, the beam is scanned in two lateral directions and many B-scans are stacked together. This volume data can be used to generate cross-sectional views, en-face images, and quantitative measurements such as retinal layer thickness.

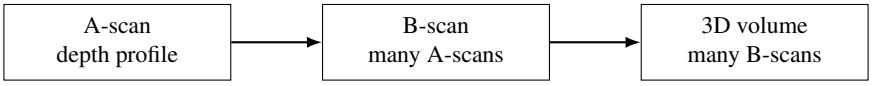


Figure 3: Formation of OCT images from A-scans, B-scans, and repeated volume scanning.

2.3 Time-Domain and Fourier-Domain OCT

In time-domain OCT, the reference mirror is mechanically moved to scan different depths. This method was important in early OCT systems, but it is relatively slow because depth information is collected sequentially.

Modern systems mainly use Fourier-domain OCT. In spectral-domain OCT, a spectrometer records the interference spectrum, while in swept-source OCT, a tunable laser rapidly sweeps through wavelengths. In both methods, Fourier transformation is used to recover the depth profile from the detected signal. These systems are faster and more sensitive than time-domain OCT [2].

Table 1: Comparison of common OCT system types.

OCT type	Basic idea	Main feature
Time-domain OCT	Reference mirror is mechanically scanned	Older and slower method
Spectral-domain OCT	Spectrometer records the interference spectrum	Faster and widely used clinically
Swept-source OCT	Tunable laser sweeps through wavelengths	High-speed and useful for deeper imaging

2.4 Resolution and Image Formation

OCT has two main types of resolution: axial and lateral. Axial resolution describes how well the system separates structures at different depths. It mainly depends on the bandwidth of the light source. For a Gaussian source spectrum, axial resolution in air can be approximated as:

$$\Delta z \approx \frac{2\ln 2}{\pi}\frac{\lambda_0^2}{\Delta\lambda} \tag{1}$$

where λ_0 is the centre wavelength and $\Delta\lambda$ is the spectral bandwidth. A broader bandwidth gives better axial resolution. Lateral resolution mainly depends on the focusing optics and beam spot size. A smaller spot improves lateral resolution, but it reduces the depth of focus, so OCT design always involves a trade-off between resolution, imaging depth, and signal strength [2].

The brightness of an OCT image represents the amount of backscattered light from tissue layers. Strongly reflecting boundaries appear brighter, while weakly scattering regions appear darker. OCT images may also contain speckle noise because the method uses coherent light.

3 System Components and Signal Processing

An OCT system includes both optical and electronic parts. The main parts are the light source, beam splitter or fibre coupler, sample arm, reference arm, scanning optics, detector, and computer processing unit. The scanning optics move the beam across the tissue so that many A-scans can be collected.

The detection method depends on the OCT type. Spectral-domain OCT uses a spectrometer and line-scan camera, while swept-source OCT uses a wavelength-swept laser and photodetector. The detected signal is not directly the final image. It must first be processed.

Common processing steps include background subtraction, resampling to linear wavenumber space, dispersion correction, Fourier transformation, and logarithmic display scaling. Fourier transformation is the key step that converts the detected interference signal into a depth profile [2].

4 Medical Applications

OCT is useful in medicine because it can show small tissue structures without cutting or removing tissue. It works best in tissues where light can enter and return enough backscattered signal. Its strongest clinical use is in ophthalmology, but it is also used in vascular imaging, dermatology, dentistry, otology, and endoscopic imaging [3].

Table 2: Selected medical applications of OCT.

Application area	Information obtained	Clinical role
Ophthalmology	Retinal layers, macula, optic nerve head, cornea	Diagnosis and follow-up
OCT angiography	Retinal and choroidal blood flow information	Non-invasive vascular assessment
Cardiology	Vessel wall, plaque, and stent structure	Intravascular imaging
Dermatology	Superficial skin microstructure	Skin lesion assessment
Dentistry	Enamel, dentine, cracks, and restorations	Non-invasive dental assessment
Endoscopy and surgery	Superficial tissue layers during procedures	Local structural guidance

4.1 Ophthalmology

Ophthalmology is the most established application of OCT. Retinal OCT can show the layered structure of the retina, making it useful for monitoring macular disease, diabetic macular oedema, retinal fluid, and glaucoma-related nerve fibre layer thinning. OCT is also used for anterior segment imaging, such as assessment of the cornea and anterior chamber angle [3].

4.2 Optical Coherence Tomography Angiography

OCT angiography, or OCTA, is an extension of OCT that detects blood flow by comparing repeated scans from the same location. It can show retinal and choroidal microvasculature without injecting dye. This is useful in conditions such as diabetic retinopathy, age-related macular degeneration, and retinal vein occlusion. However, OCTA is sensitive to motion artefacts and segmentation errors, so image quality must be checked carefully [3].

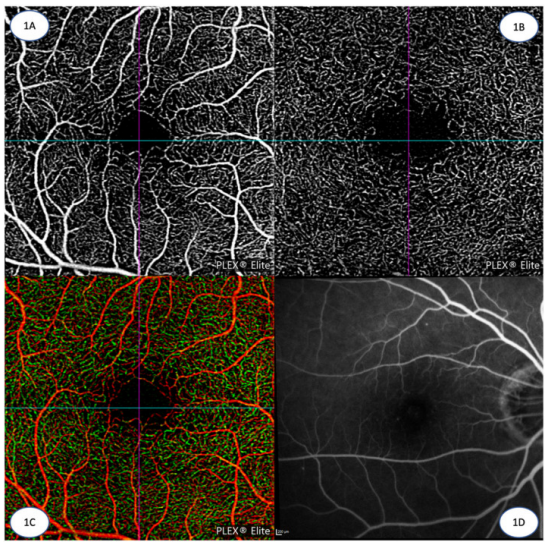


Figure 4: Example of OCT angiography showing retinal or choroidal vascular information without dye injection. Adapted from [3].

4.3 Cardiology and Intravascular OCT

In cardiology, OCT can be used inside blood vessels using a catheter-based probe. Intravascular OCT gives detailed images of the vessel wall, plaques, and stents. Its main limitation is that blood scatters light strongly, so the vessel region usually has to be cleared temporarily during imaging [3].

4.4 Dermatology

In dermatology, OCT is used to image superficial skin layers. It can help assess skin lesions, inflammatory skin diseases, wound healing, and treatment response. However, because skin scatters light, OCT is mainly useful for superficial assessment rather than deep tissue imaging [3].

4.5 Dentistry, Endoscopy, and Surgical Uses

In dentistry, OCT has been explored for detecting early caries, enamel cracks, tooth wear, and restoration defects. Endoscopic and intraoperative OCT can also provide local structural information during procedures. In these areas, OCT is best used as a complementary tool because it gives high-resolution local information but has limited penetration depth and field of view [3].

5 Quality Assurance Measures

Quality assurance is important in OCT because the final image depends on the instrument, scan settings, operator technique, patient cooperation, and image-processing software. A poor-quality scan may hide pathology, give wrong layer-thickness measurements, or create artefacts that look like disease. Therefore, OCT quality assurance should include instrument checks, patient preparation, scan acquisition, image review, documentation, and maintenance [2].

Table 3: Main quality assurance checks in OCT imaging.

QA area	What should be checked	Why it matters
Instrument performance	Resolution, sensitivity, scan depth, geometric accuracy	Ensures small structures can be seen reliably
Patient preparation	Fixation, blinking, eye movement, media opacity	Reduces poor signal and motion artefacts
Scan acquisition	Correct protocol, centering, focus, signal strength	Improves repeatability and diagnostic value
Image review	Motion, shadowing, segmentation errors, saturation	Prevents wrong interpretation
Documentation	Device, scan type, quality score, date, settings	Helps comparison during follow-up
Maintenance	Cleaning, calibration, software updates	Prevents long-term performance drift

5.1 Instrument and Performance Checks

The OCT system should be maintained according to manufacturer recommendations. Important performance checks include axial resolution, lateral resolution, sensitivity, imaging depth, geometric accuracy, and scan uniformity. These checks help confirm that small tissue layers and boundaries can be seen reliably [2].

5.2 Patient Preparation and Scan Acquisition

In ophthalmic OCT, the patient should be positioned correctly and asked to fixate steadily. Blinking, eye movement, dry eye, cataract, corneal opacity, or poor focus can reduce image quality. The operator should centre the scan on the correct anatomical region and use the appropriate scan protocol for the clinical question.

5.3 Image Quality and Artefact Checking

Every OCT image should be checked before interpretation. The operator should look for low signal strength, motion artefacts, blink artefacts, decentration, saturation, shadowing, and segmentation errors. Automated layer measurements are useful, but they should be reviewed visually when the result is used for diagnosis or follow-up.

5.4 Documentation and Maintenance

OCT reports should include the scan type, device, date, image quality score, anatomical region, and important acquisition settings. For follow-up, the same device and scan protocol should be used when possible. Regular cleaning, calibration checks, software maintenance, and operator training help keep the imaging process repeatable.

6 Limitations and Future Directions

Although OCT is very useful, it is not suitable for every medical imaging task. Its main limitation is shallow penetration depth because biological tissues scatter and absorb light. Therefore, OCT cannot replace CT, MRI, or ultrasound when deep anatomical structures need to be imaged.

OCT also has a limited field of view. It gives detailed local information, but it may not show a large region at once. Image quality can be reduced by speckle noise, motion artefacts, shadowing, poor focusing, and signal loss in deeper layers. In eye imaging, cataract or other media opacity can also reduce scan quality.

Future developments such as faster swept-source systems, OCT angiography, portable OCT devices, adaptive optics, artificial intelligence, and multimodal imaging may expand OCT into wider clinical and point-of-care applications [3].

7 Conclusion

Optical Coherence Tomography is an important medical imaging technique that uses low-coherence interferometry to create high-resolution cross-sectional images of tissue. Its main strength is that it can show micrometre-scale structures without tissue removal or ionising radiation.

OCT is most established in ophthalmology, especially for imaging the retina, macula, optic nerve head, and anterior eye structures. It is also used in OCT angiography, intravascular imaging, dermatology, dentistry, endoscopy, and surgical guidance. In these applications, OCT provides detailed local structural information that supports diagnosis, monitoring, and treatment planning.

However, OCT is limited by shallow penetration depth, limited field of view, scattering, speckle noise, motion artefacts, and dependence on good acquisition technique. Therefore, reliable OCT imaging requires proper quality assurance, including instrument checks, patient positioning, scan protocol control, artefact checking, documentation, and maintenance.

References

[1] D. Huang, E. A. Swanson, C. P. Lin, J. S. Schuman, W. G. Stinson, W. Chang, M. R. Hee, T. Flotte, K. Gregory, C. A. Puliafito, and J. G. Fujimoto, "Optical coherence tomography," *Science*, vol. 254, no. 5035, pp. 1178–1181, 1991. doi: https://doi.org/10.1126/science.1957169

[2] S. Aumann, S. Donner, J. Fischer, and F. Müller, "Optical Coherence Tomography (OCT): Principle and Technical Realization," in *High Resolution Imaging in Microscopy and Ophthalmology: New Frontiers in Biomedical Optics*, J. F. Bille, Ed. Cham: Springer, 2019. Available: https://www.ncbi.nlm.nih.gov/books/NBK554044/

[3] S. Ali, S. B. S. Gilani, J. Shabbir, K. S. Almulhim, A. Bugshan, and I. Farooq, "Optical coherence tomography's current clinical medical and dental applications: a review," *F1000Research*, vol. 10, p. 310, 2021. doi: https://doi.org/10.12688/f1000research.52031.1